VEDANT BARI

Cloud Solutions Architect | Multi-Cloud Architecture | Kubernetes | Terraform | Platform Engineering

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PROFESSIONAL EXPERIENCE

Senior Engineer

NUCLEUSTEQ | Client: Salesforce

May 2024 – Present Remote

Multi-Cloud Architecture & Platform Engineering

- Architected and implemented a reusable, YAML-driven multi-cloud infrastructure provisioning framework using Terraform and Terragrunt across AWS, GCP, and Azure.
- Designed configurable networking workflows supporting both new VPC creation and integration with existing VPCs, with subnet and CIDR configurations controlled through YAML parameters and feature flags.
- Implemented reusable Terraform modules for provisioning EKS, GKE, and AKS clusters along with their infrastructure prerequisites.
- Integrated cloud-native file storage services, including Amazon EFS, Google Cloud Filestore, and Azure Files, to support persistent data requirements for horizontally scaled applications.
- Implemented cloud-native secrets management and workload identity integrations using AWS IRSA, GCP Workload Identity, and Azure Workload Identity to provide applications with controlled access to authorized cloud resources.
- Integrated Artifactory for Docker image management and dependency caching, configuring package repository endpoints such as PyPI and npm.

Infrastructure Modernization & Security

- Coordinated and executed Amazon EKS cluster upgrade activities across Development, Integration, Pre-Production, and Production environments in collaboration with the cloud infrastructure team.
- Planned and coordinated AWS RDS MySQL upgrades from version 8.0.1 to 8.4 across Development, Integration, Pre-Production, and Production environments, including pre-migration preparation, upgrade execution, post-migration validation, and operational readiness.
- Implemented AWS EC2 Instance Metadata Service Version 2 (IMDSv2) compliance changes across EKS infrastructure in coordination with the cloud infrastructure team.
- Worked on production AWS EKS infrastructure supporting a customized OpenMetadata platform, including infrastructure operations, deployment automation, and troubleshooting.

Production Engineering & Observability

- Developed and maintained Jenkins-based CI/CD workflows for infrastructure and application deployments involving Docker and Kubernetes.
- Investigated production issues across Amazon RDS, EKS, OpenSearch, and application components using CloudWatch dashboards, Prometheus metrics, application logs, and Kubernetes events.
- Configured RDS monitoring dashboards and alarms for CPU utilization and database lock timeout conditions; used RDS Insights to investigate performance bottlenecks.

PROFESSIONAL SUMMARY

Cloud and Platform Engineering professional with nearly 8 years of experience across cloud architecture, multi-cloud infrastructure, Kubernetes, Infrastructure as Code, DevOps automation, and production engineering. Experienced in architecting and implementing reusable cloud infrastructure across AWS, Google Cloud Platform, and Microsoft Azure using Terraform and Terragrunt. Strong hands-on expertise in Kubernetes platform engineering, EKS, GKE, AKS, cloud-native storage, workload identity, secrets management, CI/CD automation, and infrastructure security. Experienced in production infrastructure modernization, database and Kubernetes upgrades, observability, troubleshooting, and resource optimization.

EDUCATION

Bachelor of Engineering — Computer Engineering

Ramrao Adik Institute of Technology, Nerul

University of Mumbai

September 2013 – May 2017

CERTIFICATIONS

Google Cloud Professional Cloud Architect

AWS Certified Solutions Architect – Associate

AWS Certified Developer – Associate

Certified Kubernetes Application Developer (CKAD) — The Linux Foundation

AWARDS & ACHIEVEMENTS

Valuable Team Player Award (2021):
Recognized for contributions to team performance and project delivery.

Second Prize — National-Level Students Conference on Frontiers in Engineering and Technology (2017):
Presented a paper on an Automated Fire Detection and Extinguishing Bot.

CORE TECHNICAL SKILLS

Cloud Platforms: AWS, Google Cloud Platform (GCP), Microsoft Azure

Cloud Architecture: Multi-Cloud Architecture, Infrastructure Design, High Availability, Scalability, Cloud-Native Platforms, Infrastructure Standardization, Resource Optimization

- Improved monitoring and alerting workflows using Prometheus, Grafana, Alertmanager, Splunk, Loki, and AWS CloudWatch.
- Contributed to Monitor-as-Code migration by moving monitoring configurations from UI-based setup to code, validating YAML configurations, importing monitors, and automating deployment through Jenkins.
- Investigated production indexing-related performance issues in OpenMetadata and supported operational monitoring improvements.
- Contributed to infrastructure and container optimization initiatives that reduced resource consumption by approximately 25%.

Solutions Developer

TATA TECHNOLOGIES | Client: Tata Motors

December 2021 – October 2023 Thane, Maharashtra

- Administered and maintained Kubernetes clusters on Amazon EKS, supporting application availability, scalability, and deployment operations.
- Automated Kubernetes application deployments using Jenkins pipelines to improve deployment consistency and operational efficiency.
- Implemented Kubernetes security practices, including RBAC, network policies, and encryption.
- Collaborated with application and infrastructure teams to design and troubleshoot scalable deployment solutions.
- Investigated Kubernetes workload, networking, and application deployment issues.
- Provided technical guidance to team members on Kubernetes fundamentals, deployment practices, and troubleshooting.

Software Developer

NETWORK INTELLIGENCE INDIA PRIVATE LIMITED

March 2020 – October 2021 Thane, Maharashtra

- Developed backend services and REST APIs using Python and Django REST Framework.
- Created Jenkins-based continuous delivery pipelines, including Docker image build workflows.
- Supported AWS-hosted application infrastructure and containerized application deployments.
- Implemented asynchronous background processing using Celery and RabbitMQ and scheduled recurring tasks using Celery Beat.
- Integrated backend APIs with Angular frontend components and supported application debugging and maintenance.

Python Backend Developer

ANALOGY PLUS PRIVATE LIMITED

May 2018 – February 2020 Thane, Maharashtra

- Developed backend APIs and application functionality using Python and Django REST Framework.
- Containerized applications using Docker to support consistent development and deployment environments.
- Implemented Redis caching to improve application response performance.
- Worked with PostgreSQL for application data storage and management.
- Integrated Firebase Cloud Messaging for application notification functionality.

Containers & Orchestration: Kubernetes, Amazon EKS, Google Kubernetes Engine (GKE), Azure Kubernetes Service (AKS), Docker, Helm, Istio

Infrastructure as Code: Terraform, Terragrunt, YAML-driven Infrastructure Configuration

Cloud Networking & Security: VPC, Subnets, CIDR, IAM, IRSA, GCP Workload Identity, Azure Workload Identity, IMDSv2

Storage & Secrets: Amazon EFS, Google Cloud Filestore, Azure Files, AWS Secrets Manager, Cloud-native Secrets Management

CI/CD & Artifact Management: Jenkins, CI/CD Pipelines, JFrog Artifactory, Docker Registry, PyPI, npm

Monitoring & Observability: Prometheus, Grafana, Alertmanager, Splunk, Loki, AWS CloudWatch

Databases & Platforms: Amazon RDS, OpenSearch, PostgreSQL, OpenMetadata

Programming & Scripting: Python, Bash/Shell, Django REST Framework

SELECTED CLOUD ARCHITECTURE & PLATFORM PROJECTS

1. Multi-Cloud Kubernetes Infrastructure Blueprint

AWS | GCP | Azure | Terraform | Terragrunt | YAML | EKS | GKE | AKS

- Designed and implemented a reusable infrastructure provisioning framework supporting multiple cloud providers through configurable YAML inputs.
- Enabled flexible networking configurations supporting both new and existing VPCs, including client-specific subnet and CIDR requirements.
- Integrated Kubernetes provisioning, cloud-native storage, secrets management, and workload identity to support application deployment requirements.
- Standardized environment-specific infrastructure provisioning using Terraform modules and Terragrunt orchestration.

2. AWS EKS and OpenMetadata Platform

AWS EKS | Kubernetes | Terraform | Docker | Jenkins

- Supported production infrastructure for a customized OpenMetadata platform running on Amazon EKS.
- Worked on infrastructure operations, deployment automation, Kubernetes troubleshooting, and platform maintenance.
- Supported monitoring and investigation of infrastructure and application-level issues.

3. Cloud Infrastructure Modernization

AWS | Amazon RDS | EKS | IMDSv2 | CloudWatch

- Coordinated RDS MySQL and EKS upgrade activities across Development, Integration, Pre-Production, and Production environments.
- Implemented IMDSv2 compliance changes across EKS infrastructure.
- Supported database monitoring, upgrade validation, and production infrastructure troubleshooting.